Lesson 2.9 Guided Notes

Session 1 · Finding Mean Absolute Deviation (MAD)

I can find the mean absolute deviation of a data set and explain what it measures. · 6.SP.B.5c

Word / Spanish	Meaning — complete the blank	Example
mean absolute deviation (MAD) / desviación media absoluta	The _____ of the distances from the mean.	distances 4, 0, 1, 3: MAD 2
deviation / desviación	How _____ one value is from the mean.	mean 19, value 23: 4
absolute value / valor absoluto	A number's distance from 0; it is never _____.	|−5| = 5
spread / dispersión	How far _____ the values are.	small MAD = tight set

Key idea

Find the mean, then each value's _____ from the mean.

The MAD is the _____ of those distances.

A small MAD means the values are _____ to the mean.

Worked example

Four plants grew 3, 7, 8 and 10 cm. Find the MAD.

Mean: 28 ÷ 4 = _____

Distances from 7: 4, _____, 1, 3

Add the distances: _____. Divide by 4.

Answer: MAD = _____ cm

You try

1. The mean is given. Find the MAD.
a. 2, 4, 6, 8 (mean 5) _____
b. 8, 14, 16, 22 (mean 15) _____

_____

2. The mean of 4, 6, 8, 10, 12 is 8. Find the MAD.

_____

Sum it up: The MAD tells me ___ because ___.

_____

Session 2 · Comparing Data Sets Using MAD

I can compare two data sets that share a mean by comparing their mean absolute deviations. · 6.SP.B.5c

Word / Spanish	Meaning — complete the blank	Example
variation / variación	How _____ the values are from each other.	larger MAD = more variation
consistent / consistente	Values stay _____ together; the spread is small.	small MAD = consistent
MAD / desviación media absoluta	The mean _____ of the values from the mean.	MAD 1.5 versus MAD 4

Key idea

Two sets can have the same mean and _____ MADs.

The set with the _____ MAD is steadier (more consistent).

Worked example

Baker A made 8, 9, 11, 12 cookies per tray. Baker B made 4, 8, 12, 16. Both means are 10. Who is steadier?

A's distances: 2, 1, 1, 2. MAD = 6 ÷ 4 = _____

B's distances: 6, 2, 2, 6. MAD = 16 ÷ 4 = _____

Answer: Baker _____ is steadier: 1.5 < 4.

You try

1. Both sets have a mean of 30. Find each MAD.
a. Set 1: 26, 28, 32, 34 _____
b. Set 2: 20, 24, 36, 40 _____

_____

2. Machine A fills bottles with mean 500 mL and MAD 2 mL. Machine B: mean 500 mL, MAD 14 mL. Which is more consistent?

_____

Sum it up: When two sets have the same mean, I compare ___ because ___.

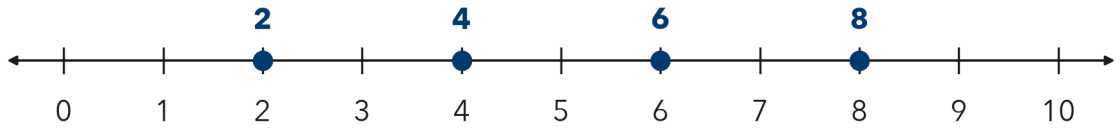
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Session 1 · Model, reason, and explain

Finding Mean Absolute Deviation (MAD) · 6.SP.B.5c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Plant growth measurements are 2, 4, 6, and 8 centimeters. Find the mean, each distance from the mean, and the MAD. Explain why the distances do not cancel.



_____

_____

_____

Extend your reasoning

Explain the difference between the mean of 5 cm and MAD of 2 cm in one connected sentence.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Comparing Data Sets Using MAD · 6.SP.B.5c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A: 4, 5, 5, 6 and B: 1, 3, 7, 9 have the same mean. Compute their MADs and explain which set is more consistent around its mean.

A: 4, 5, 5, 6

B: 1, 3, 7, 9

_____

_____

_____

Extend your reasoning

Create four observations with mean 5 and MAD 0. Explain your set.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?